



Soil–wheat nutrient interactions following fertilization with digestate from one- and two-stage molasses anaerobic digestion: a field study

Marta Galas · Anna Detman-Ignatowska · Jerzy Jonczak ·
Joanna Szewińska · Aleksandra Chojnacka

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Abstract

Aims Sustainable soil management besides ecologically friendly alternatives to mineral fertilizers, requires also bioconversion technologies capable of generating soil amendments with targeted agronomic functions. Two-stage system of anaerobic digestion with spatial separation of processes into two bioreactors allows to obtain two-stage digestate (2-SD), which possibly differs from conventional one-stage digestate (1-SD). This study evaluated whether these process-derived differences translate into fertilizing properties of digestates, altered soil functioning and plant responses.

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M. Galas (✉) · J. Szewińska · A. Chojnacka
Department of Biochemistry and Microbiology, Faculty of Biology and Biotechnology, Warsaw University of Life Sciences (SGGW), Nowoursynowska 159, 02-776 Warsaw, Poland
e-mail: marta_galas@sggw.edu.pl

A. Detman-Ignatowska
Laboratory of White Biotechnology, Institute of Biochemistry and Biophysics, Polish Academy of Sciences, Pawińskiego 5a, 02-106 Warsaw, Poland

J. Jonczak
Department of Soil Science and Agricultural Chemistry, Faculty of Bioeconomy and Food Systems, Warsaw University of Life Sciences (SGGW), Nowoursynowska 159, 02-776 Warsaw, Poland

Methods A field experiment with wheat comparing 2-SD digestate with 1-SD digestate, synthetic fertilization (SF), referred to unfertilized control (NF) was performed. Soil physicochemical properties, plant nutrient status, digestate chemistry and microbial consortia were analyzed. Soil–plant interactions were assessed through root nutrient transporters genes expression and finally crop yield was determined.

Results Compared with 1-SD, 2-SD digestate exhibited a higher mineralization degree and distinct microbial diversity, indicating different chemical and microbiological functionality. Only bioavailable phosphorus was increased in soil treated with 1-SD digestate. Both digestates treatments enhanced root *TaAMT1.2* gene expression, altering nitrogen acquisition pattern. 2-SD digestate treatment caused stronger yield response than 1-SD (2.8 t ha⁻¹ vs 2.1 t ha⁻¹), while both maintained soil pH stability unlike SF. Crop performance was slightly associated with soil pH and *TaPHT1.10* gene expression.

Conclusions Findings demonstrate that 2-SD digestate is a functionally distinct, more mineralized, biologically active soil amendment from 1-SD digestate. Both with specific effects on soil chemistry, plant nutrient uptake patterns and enhanced productivity, highlighting two-stage anaerobic digestion as a promising source for circular nutrient management in cereal tillage.

Keywords Soil amendment · Ion transporter · *TaAMT1.2* transporter · Digestate · Wheat · Crops

Introduction

Modern agriculture faces many challenges, particularly in cereal crop production, which provides the primary staple food for most people. Among cereals, wheat (*Triticum aestivum* L.) is one of the most widely cultivated species, valued both as a food source and for industrial purposes (Vitousek et al. 2009; Duan et al. 2014; Salehi et al. 2025). However, wheat has high nutritional demands and requires fertile soil for productive cultivation. Additionally, a continuously growing global population places a premium on increasing crop yields, fostering excessive agricultural intensification. Consequently, modern agricultural systems remain strongly dependent on external nutrient inputs (Hlisenikovsky et al. 2023; Salehi et al. 2025). Synthetic fertilizers, which mainly boost soil nitrogen and phosphorus content in mineral forms available for plant uptake, have been widely used in agriculture for decades. Despite their positive impact on crop yields, their excessive use is not economically justified and can also pose a threat to the natural environment. Long-term application of mineral fertilizers, especially those with high soluble ammonium sulfate content, contributes to soil acidification, alters soil microbiome communities (Howe et al. 2024), and increases nitrogen and phosphorus leaching, leading to eutrophication and degradation of agroecosystems (Reijnders 2008; Morsetto 2020; Guo et al. 2023; Salehi et al. 2025). These challenges highlight the urgent need for alternative fertilization strategies that sustain crop productivity while reducing negative environmental impacts. Thus, developing natural, biological approaches to crop fertilization, particularly those involving microorganisms, appear to be a safer option. Among these, digestate, the final product of the anaerobic digestion (AD) process, has recently gained considerable attention.

Digestate is an organic matter-rich substance that represents a unique type of biofertilizer because it combines substantial amounts of organic carbon, as well as nitrogen, phosphorus, often in forms that can be gradually released into the soil (Engedal et al. 2025), with diverse complexes of microbial consortia. This integration of both chemical and biological functions is highly desired in sustainable agriculture and promote soil health (Engedal et al. 2025), posing digestate as a biologically active amendment. It is rich in microorganisms originating from AD processes,

which may influence native soil microbial activity and promote nutrient cycling (Fernandez-Bayo et al. 2020; van Midden et al. 2023). This microbial consortium is particularly important, as an increasing number of studies have indicated that the beneficial effects of biofertilizers are more often driven by interactions within microbial communities rather than by individual strains applied alone (Chen et al. 2006; Malboobi et al. 2009; Raimi et al. 2020; Cheng et al. 2023). Microbial consortia may significantly enhance organic matter mineralization, phosphorus solubilization (Chen et al. 2006; Pankiewicz et al. 2019; Pan and Cai 2023) and the modulation of rhizosphere microbial processes, particularly those involved in nitrogen cycling in soil (Guo et al. 2023). These properties distinguish the use of digestate in sustainable agriculture from the application of single-strain biofertilizers or traditional mineral fertilizers. This perspective is especially relevant because it aligns with circular economy practices, in which waste from one biotechnological process becomes a substrate for another reaction or is repurposed as a resource of economic value, thereby maximizing resource utilization (Morsetto 2020; Chojnacka et al. 2019; Jin et al. 2022). During sugar production from beet roots, molasses is generated as a by-product. Molasses is a high-energy organic compound rich in readily fermentable carbon, making it an attractive substrate for AD (Fang et al. 2011). It is used in both one-stage (traditional) and two-stage systems (Chojnacka et al. 2015; Detman et al. 2017).

The two-stage (2-SD) system of anaerobic substrate decomposition differs fundamentally from the conventional one-stage (1-SD) process by separating hydrolysis and acidogenesis from acetogenesis and methanogenesis across two distinct bioreactors (Chojnacka et al. 2026). Moreover, the technological separation of microbial processes substantially modifies digestate composition, including nutrient availability, organic carbon fractions, and microbial community structure (Chojnacka et al. 2026). To date, digestates obtained from one-stage AD have been relatively well characterized as soil amendments and biofertilizers, with numerous studies reporting their potential to improve soil fertility and crop productivity (Slepetiene et al. 2020; Zhao et al. 2024; Engedal et al. 2025; Zapata-Morales and Moreno-Andrade 2025). By contrast, the fertilizing properties of digestates derived from two-stage AD systems remain poorly

understood. Given that their composition differs from the digestates obtained from one-stage AD process, it cannot be assumed that they behave similarly; this knowledge gap is particularly evident in cereal crop research.

In this study, we compared the effects of one-stage (1-SD) and two-stage (2-SD) digestates applied as biofertilizers under field conditions. Digestates represent nutrient-rich organic amendments; however, from a plant physiological perspective, it is important to emphasize that roots can readily absorb nutrients only present in plant-available forms in the soil, predominantly mineral (inorganic). As digestates also exhibit a high mineral content, their application as biofertilizers appear to offer an advantage in wheat cultivation (Simon et al. 2015; Barłóg et al. 2020; Winkhart et al. 2024). Ion uptake in roots occurs via specialized affinity protein transporters, whose activation depends on ion concentrations in the soil (Duan et al. 2015; Zhang et al. 2014; Liu et al. 2015). Those protein ion transporters are encoded by genes in roots. Subsequently, we investigated expression of: *TaAMT1.2* gene encoding high-affinity transporter of ammonium (Liu et al. 2015; Duan et al. 2015), *TaNRT1.1* gene encoding dual-affinity nitrate ions transporter (Duan et al. 2015; Taulemesse et al. 2015; Fang et al. 2021) and eventually, *TaPHT1.10* gene (formerly known as *TaPT2* or *TaPHT1.10-U*), which encodes high-affinity phosphates ion transporter (Zhang et al. 2014; Teng et al. 2017). Linking ion transporter gene expression with soil chemical composition may therefore demonstrate the plant's physiological response to treatment in nutrient acquisition at the end of the growing season to assess whether nutrients uptake derived from digestates are not lost too rapidly.

Based on this information, we hypothesized that: (i) digestates obtained from one-stage and two-stage AD differ in their chemical stoichiometry and microbial community structure, resulting in distinct effects on soil nutrient availability under field conditions; (ii) digestate-based fertilization modifies wheat nutrient acquisition mechanisms, particularly by change in the expression of ion transporters in roots. We further hypothesized that: (iii) both digestates treatments modify nitrogen acquisition mechanisms pattern in wheat roots despite equal input of this nutrient at fertilizing dose to synthetic mineral treatment.

Integrating interdisciplinary approaches described below, to our knowledge, this study provides the first field-based evaluation of the fertilizing properties of valuable waste product—digestates derived from two-stage AD of molasses, while explicitly comparing one-stage (traditional) and two-stage digestates with mineral fertilization. This work directly contributes to the development of sustainable fertilization strategies within circular-economy frameworks and to the needs of modern ecological agriculture.

Materials and methods

Experiment design under field conditions and fertilization regime

The experiment was conducted under environmental conditions in randomized field microcosms from September 2023 until the end of the vegetative season in July 2024. Two hundred winter wheat (*Triticum aestivum* L.) 'Euforia' grains were sown in each microcosm with total area 0.5 m² (corresponding to the recommended density of 400 grains per m²), following a randomized block design with five replicates per treatment (Fig. 1B). The experimental variants included (i) NF: non-fertilized plants (control), (ii) SF: fertilization with the multi-component synthetic mineral fertilizer Azofoska, (iii) 1-SD: fertilization with one-stage anaerobic digestate, and (iv) 2-SD: fertilization with two-stage anaerobic digestate.

To obtain one-stage and two-stage digestates, molasses (20 g L⁻¹) was initially subjected to AD, according to the method described by Detman et al. (2017) and Chojnacka et al. (2026), as schematically presented in Fig. 1A. The fertilizer was at a final dose of 140 kg ha⁻¹ N (Kotwica et al. 2011). Liquid digestates derived from one-stage (1-SD) and two-stage (2-SD) AD systems were applied to meet the nitrogen demand of plant mineral nutrition, analogously to the mineral fertilization used in the SF treatment (Azofoska, Poland; chemical composition: 13.6% total nitrogen (N) [5.5% of nitrate nitrogen + 8.1% of ammonium nitrogen], 6.4% phosphorus pentoxide (P₂O₅) soluble in neutral ammonium citrate solution and water, 19.1% potassium oxide (K₂O), 4.5% magnesium oxide (MgO), 23.0% sulfur trioxide (SO₃), 0.045% boron (B), 0.8% copper (Cu), 0.17% iron

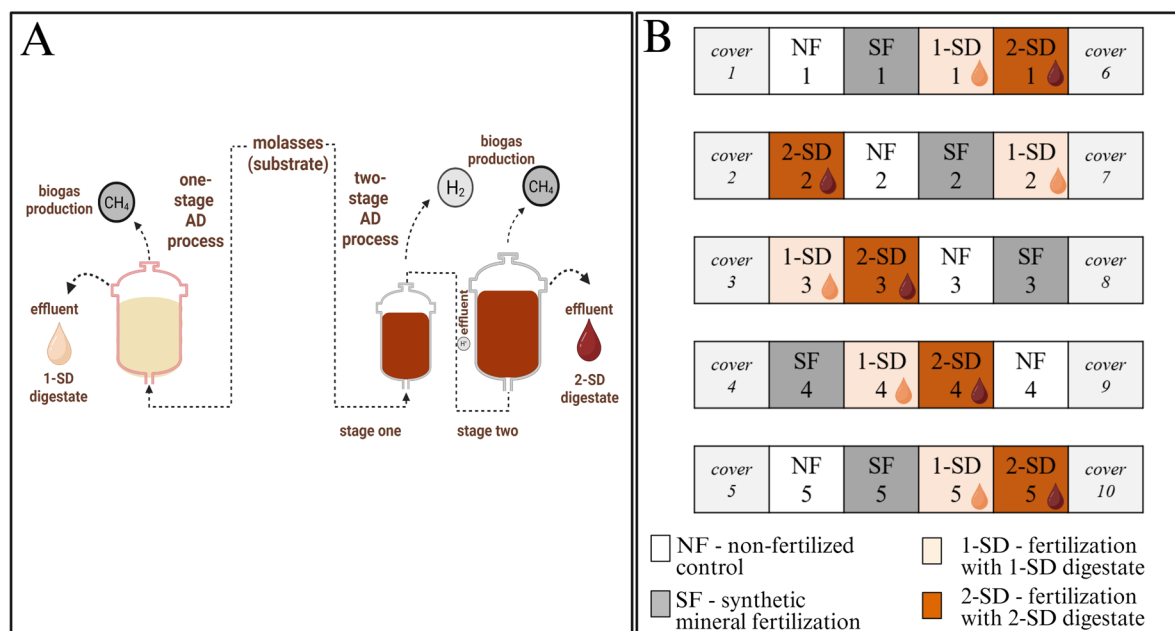


Fig. 1 Schematic design of digestate production and its utilization as biofertilizer. **A** Illustrates the anaerobic digestion process in one-stage and two-stage systems, leading to the production of 1-SD and 2-SD digestates. **B** Shows the field experiment design. The outer gray rectangles represent cover microcosms sown with wheat to minimize the influence of other

environmental effects on the experimental plots; these were not included in the analysis. The inner rectangles represent the experimental plots. White boxes indicate NF control variants and gray boxes indicate SF variants. Orange boxes indicate 1-SD fertilization variants, and light brown boxes indicate 2-SD fertilization variants using liquid digestates

(Fe), 0.27% manganese (Mn), 0.04% molybdenum (Mo), and 0.045% zinc (Zn)).

Fertilization was applied twice at the same dose during the vegetative season, in early spring (March, in BBCH 25 wheat stage) and six weeks later (during BBCH 31 wheat stage), according to Zadoks et al. (1974) scale of cereal growth stages. Wheat cultivation continued until full grain maturity, after which the crops were harvested at the beginning of August. Border microcosms were also sown with wheat to minimize edge effects; however, they were not included in the analysis, as represented in Fig. 1B.

Soil sampling and chemical analyses

Soil samples were collected before the start of the experiment and after crop harvest. Three soil subsamples were taken from each plot at a depth of 0–30 cm (humus horizon) using a 3-cm soil corer, then combined to form a single composite sample. The samples were cooled and transported to the laboratory immediately after collection. Fresh samples were

subjected to extraction using a 1% potassium sulfate solution, and the contents of $\text{NH}_4\text{-N}$ and $\text{NO}_3\text{-N}$ in the extracts were determined using Nanocolor No. 91865 and Spectroquant No. 1147520001 tests, respectively, with a Nanocolor VISII spectrophotometer. Soil moisture content was determined in each sample using the gravimetric method at temperature of 105 °C. The remaining portions of the samples were air-dried and sieved through the 2.0-mm sieve to remove coarse fragments. Subsequent analyses of the <2.0-mm (earth) fraction were performed, a portion of which (~10 g) was milled into a powder using the SPEX SamplePrep ball mill equipped with zirconium milling chamber. Final size of soil particles is <10 µm according to producer data. Laboratory analyses included determination of particle-size distribution using pipette and sieving methods, with classification of textural fractions and groups according to the Polish Soil Science Society (PTG, 2009), which is compatible with USDA standards. Soil pH was determined via a potentiometric method (SevenDirect SD23, Mettler Toledo, Switzerland) in

suspensions with water (pH-H₂O) and 1 mol L⁻¹ KCl solution (pH-KCl), using a soil-to-water ratio of 1:2.5. TOC and TN contents were determined via dry combustion (Vario MacroCube, Elementar, Germany). Total phosphorus (P_T) was measured via inductively-coupled plasma atomic emission spectrometry (ICP-AES; Avio 200, Perkin Elmer, USA) following sample digestion in a mixture of 65% nitric acid, 38% hydrochloric acid, and 40% hydrofluoric acid (3:2:5, v/v/v) using a microwave digestion system (EthosUp, Milestone, Italy). Bioavailable phosphorus (P_B) via ICP-AES (Avio 200, Perkin Elmer, USA) after extraction according to the Mehlich III procedure (Mehlich 1984).

Chemical characterization of digestates

The chemical characterization of digestates included the determination of TOC and TN via dry combustion (Vario MacroCube, Elementar, Germany) and P_T via ICP-AES (Avio 200, Perkin Elmer, USA), following sample digestion in 65% nitric acid using a microwave digestion system (EthosUp, Milestone, Italy). Mineral forms of nitrogen (NH₄-N and NO₃-N) were determined using the NANOCOLOR tests (Ammonium 200 ISO 7150-1, DIN 38406-E5, EPA 350.1, and NANOCOLOR® Nitrate 50 ISO 7890-1, DIN 38405-D9, ISO 23696-1). Digestate pH was measured using an ELMETRON CP-502 pH meter equipped with an IJ44C pH electrode (ELMETRON, Poland).

Microbiological characterization of digestates

Genomic DNA from both one-stage and two-stage digestates was isolated using the *DNeasy PowerSoil Pro Kit* (cat. no. 47014, QIAGEN) according to the manufacturer's protocol. DNA quality and quantity were verified via spectrophotometry and agarose gel electrophoresis. Bacterial and fungal community composition was characterized using targeted amplicon sequencing of the bacterial 16S rRNA gene (V3–V4 hypervariable region) and the fungal internal transcribed spacer (ITS) region, respectively. The V3–V4 region was amplified using the universal bacterial primer pair 341F (5'-CCTACG GNGGCCWGCAG-3') and 806R (5'-GGACTA CHVGGGTWTCTAAT-3'). The fungal ITS region was amplified using the primer pair ITS1F (5'-CTT

GGTCATTTAGAGGAAGTAA-3') and ITS2 (5'-GCTGCGTTCTTCATCGATGC-3'). PCR amplification was performed using a high-fidelity DNA polymerase. Amplicons were purified, indexed, and sequenced in paired-end mode on Illumina platforms. Raw sequencing reads were demultiplexed and processed using a standardized bioinformatics workflow. Adapter and primer sequences were removed, and reads were quality-filtered to exclude low-quality bases. Denoising, paired-end merging, and chimera removal were performed to generate amplicon sequence variants (ASVs). Taxonomic assignment of 16S rRNA gene ASVs was conducted against the SILVA ribosomal RNA gene reference database, while fungal ITS ASVs were classified using the UNITE database for fungal internal transcribed spacer sequences. All data collected during sequencing has been uploaded into NCBI repository with BioProject accession number PRJNA1423286.

Analysis of ion transporter gene expression

RNA isolation from wheat roots collected at harvest (in August) was performed according to the CTAB method described by Wang and Stagemann (2010), with modifications (Rucińska et al. 2021; Wang et al. 2022). Roots were washed three times with sterile Mili-Q water to remove residual soil from the tissue. Next, 150 mg of wheat root tissue was frozen in liquid nitrogen and ground using a sterile mortar. Samples were subjected to lysis using 1 mL of CTAB-based extraction buffer (EurX sp.z o.o, Poland) at a final pH of 8.0. The buffer contained 2% (w/v) CTAB, 20 mM EDTA, 1.4 M NaCl, and 100 mM TRIS, with β-mercaptoethanol (Sigma-Aldrich, USA) added to a final concentration of 1% (v/v) immediately before use. The isolated nucleic acids were dissolved in 100 μL of TE buffer (10 mM TRIS, 1 mM EDTA; pH 8.0) from the *DNeasy PowerSoil Pro Kit* (cat. No. 47014, QIAGEN) and treated with DNase (A&A Biotechnology) according to the manufacturer's instructions. RNA quality and quantity (A_{260/280} and A_{260/230} absorbance ratios, respectively) were measured using a NanoPhotometer (NP80, IMPLLEN) and verified via electrophoresis on a 2% agarose gel. Samples were used as a template for cDNA synthesis in a total reaction volume of 20 μL using the QuantiTect Reverse Transcription Kit (QIAGEN, cat. No. 205311). The procedure was performed according

to the manufacturer's protocol. The final cDNA concentration after reverse transcription was adjusted to 20 ng μL^{-1} for subsequent molecular analyses.

Gene expression analysis of ion transporters was performed using digital PCR (dPCR) (Voegel et al. 2021; Morcia et al. 2020; Shynggyskyzy et al. 2025). Reactions were carried out on a QIAGEN QIAcuity One 5-Plex system using EvaGreen Mastermix®. For each reaction, 8.5 k nanoplate formats were used. Primer pairs and expected amplicon sizes were first validated to ensure optimal reaction conditions. For each sample, the cDNA template was diluted tenfold or 100-fold, as appropriate. Reactions were carried out in a final volume of 12 μL using Eva Green Mastermix® and 10 μM forward and reverse primers. The genes encoding ion transporters in wheat roots analyzed in this study are listed in Table 1. Additional information about the dPCR reaction conditions is available in an open data repository (<https://doi.org/10.18150/7ZYVRV>).

Plant material collection, chemical analyses, and wheat yield determination

All plants were harvested separately from all plots and air-dried. Next, the total biomass of grain and straw was determined for each plot, and then this

was converted to tons per hectare. Grain and straw samples were milled into a powder and subjected to chemical analysis. TOC and TN contents were determined via dry combustion (Vario MacroCube, Elementar, Germany), while P_T was determined via ICP-AES (Avio 200, Perkin Elmer, USA), following sample digestion in 65% nitric acid using a microwave digestion system (EthosUp, Milestone, Italy).

Statistical analysis

Data analysis was performed in R (version 4.0.3). The Shapiro–Wilk test was applied to check whether the data followed a normal distribution. If data followed normal distribution, a t-test or one-way ANOVA, followed by Tukey's multiple comparison test, was conducted. When the data did not meet the criteria for normal distribution, the Mann–Whitney U test or nonparametric Kruskal–Wallis rank sum test was performed using the R-Commander package, followed by Dunn's post-hoc test. Linear regression analyses were conducted using GraphPad Prism (version 10.6.0). Figures were prepared using BioRender software. The significance level for all tests was set at $\alpha=0.05$. All raw data used for statistical analyses have been deposited in an open data repository (<https://doi.org/10.18150/7ZYVRV>).

Table 1 Genes encoding ion transporters in wheat roots whose expression profiles were analyzed

Target gene	Ions transported by encoded gene	Sequence of used primers	Annealing temperature	Reference
<i>TaAMT1.2</i>	Ammonium	Forward	5' TGA GGT AGG CGA CGA ACT GC 3'	57–55°C, modified in this study (Duan et al. 2015)
		Reverse	5' CAA CGG GTT TAT CGG GAA GC 3'	
<i>TaNRT1.1</i>	Nitrates, nitrites	Forward	5' TTG TAG AGC AAG GGA TGG 3'	48–49°C, modified in this study (Duan et al. 2015)
		Reverse	5' TTC GCT GTG GTG ATG TAA T 3'	
<i>TaPHT1.10</i>	Phosphates	Forward	5' GCG TTC GGG TTC CTG TAT GC 3'	58–60°C Teng et al. 2017; Duan et al. 2015)
		Reverse	5' CAG TCG GAG CAA TGG TGT CGT 3'	
<i>GAPDH</i>	Positive control	Forward	5' GAC TCC TCC TTG ATA GCA GCC TT 3'	52°C (Duan et al. 2015)
		Reverse	5' CAA CGC TAG CTG CAC CAC TAA CT 3'	

Primer sequences were selected according to relevant literature sources. The table also contains any modifications implemented to optimize dPCR reaction conditions

Results

Soil chemical properties

The soil used in this experiment exhibited a sandy texture, with 82.2% sand, 16.2% silt, and 1.6% clay. The initial chemical composition of the soil is presented in Table 2, including the ranges of total element content and their bioavailable forms, to show the changes over time following fertilization treatments in the SF, 1-SD, and 2-SD microcosms. Initial mean soil pH was around 7.35. Post-experiment chemical analyses revealed a statistically significant increase in P_B content in soil samples from the 1-SD digestate treatment, with approximately 23.3% higher content compared with the NF control (Tukey's post-hoc test; $p < 0.05$) (Table 2). Additionally, a statistically

significant slight decrease in pH was observed in SF soil samples compared to the NF control variants (Table 2). No significant differences were observed in other soil parameters, including TOC, TN, NH_4-N , NO_3-N contents, and the TOC:TN ratio.

Chemical characteristics of digestates

To assess whether the application of digestates as biofertilizers was efficient and whether soil parameters were affected during the experimental treatment, chemical analyses of element content and pH were conducted on 1-SD and 2-SD digestates derived from the anaerobic decomposition of molasses. The results are shown in Table 3. The 1-SD digestate exhibited 24.6% higher TN concentrations (t-test; $p < 0.001$) and 19.7% higher P_T content (t-test; $p < 0.001$) than

Table 2 Elemental composition of soil

		Initial Soil Before Experiment	Soil After Experiment			
			NF	SF	1-SD	2-SD
TOC	Content [mg kg^{-1}]	13259.8±345.6	13859.3±177.3 ^a	13779.3±663.7 ^a	13617.8±427.3 ^a	13493.5±388.8 ^a
TN		1273.5±23.8	1323.5±89.8 ^a	1301.9±98.9 ^a	1319.7±45.7 ^a	1278.4±65.9 ^a
NH_4-N		8.95±1.17	7.84±0.38 ^a	8.36±0.84 ^a	8.64±0.61 ^a	8.25±0.77 ^a
NO_3-N		1.58±0.16	11.9±2.83 ^a	15.1±3.05 ^a	14.0±2.61 ^a	12.5±2.66 ^a
P_B		297.4±28.9	330.3±30.1 ^b	347.1±21.6 ^{ab}	407.5±53.2^a	381.0±22.7 ^{ab}
TOC:TN		10.4±0.18	10.5±0.66 ^a	10.6±0.76 ^a	10.3±0.63 ^a	10.6±0.53 ^a
pH		7.35±0.07	7.35±0.08 ^a	7.08±0.16^b	7.39±0.10 ^a	7.38±0.13 ^a

Mean ± SD values of total element content, their mineral forms, associated ratios, and pH for the initial soil used in the experiment are represented in the left side of the table. For soil samples collected after the experiment and wheat harvest ($n=5$), total element content of selected nutrients, their mineral ion forms associated ratios, and pH (mean ± SD), are shown on the right side of the table. One-way ANOVA was conducted, followed by Tukey's post-hoc comparison. Statistically significant differences ($p < 0.05$) are indicated in bold and marked with different statistical letters in superscript next to result in the table

Table 3 Comparison of selected element content and elemental ratios in 1-SD and 2-SD digestates ($n=5$), as well as pH values ($n=102-109$)

	TOC	TN	P_T	TOC:TN	TOC: P_T	TN: P_T	NH_4-N	NO_3-N	pH
	[mg kg^{-1}]						[mg L^{-1}]		
1-SD	1246.6	1002.3	1121.8	1.33	1.10	0.90	298.8	29.9	6.72
	± 414.3 ^a	± 162.8^a	± 74.5^a	± 0.63 ^a	± 0.31^a	± 0.20 ^a	± 23.1^a	± 5.93^a	± 0.63^a
2-SD	1231.7	755.9	901.4	1.67	1.36	0.84	370.1	41.8	6.87
	± 290.2 ^a	± 87.7^b	± 75.9^b	± 0.47 ^a	± 0.27^b	± 0.13 ^a	± 59.4^b	± 6.36^b	± 0.14^b

Values show mean ± SD. Statistical analyses were performed using t-tests or Mann–Whitney U tests, depending on normality of data distribution. Statistically significant differences ($p < 0.05$) are marked with different statistical letters and bolded in the table

the 2-SD digestate (Table 3). A higher TOC:P_T ratio was observed in the 2-SD digestate (Mann–Whitney *U* test; $p < 0.05$), although no significant difference in TOC:TN ratio was detected between digestates. 2-SD digestate had higher mineralization status than 1-SD digestate, which was investigated by 23.9% higher NH₄-N (Mann–Whitney *U* test, $p < 0.05$) and 39.8% higher NO₃-N (t-test; $p < 0.05$) concentrations in 2-SD than 1-SD digestate, respectively. Significant differences in pH were also noted, with greater alkalization observed in the 2-SD digestate (Mann–Whitney *U* test; $p < 0.001$) (Table 3). No statistically significant differences in the TN:P_T ratio were found between 1-SD and 2-SD digestates (Table 3).

Microbial composition of digestates

High-throughput sequencing analysis revealed distinct microbial community profiles in the 1-SD and 2-SD digestates obtained from AD of molasses (Fig. 2). Differences were observed in both bacterial

and fungal assemblages, reflected by variation in the relative abundance of dominant taxa between the two digestates. The microbial community of the 1-SD digestate was characterized by a higher relative abundance of bacterial genera associated with the fermentative and acidogenic stages of AD. Dominant bacterial taxa included *Lactiplantibacillus*, *Bacteroides*, *Ruminococcus*, *Eisenbergiella*, *Lactococcus*, *Macellibacteroides*, and *Enterobacter* (Fig. 2A). In addition, *Methanobacterium* and *Spaerochaeta* were detected at moderate relative abundance (Fig. 2A). The fungal community in the 1-SD digestate was dominated by *Trichoderma*, with additional contributions from *Mortierella*, *Penicillium*, *Cladosporium*, and *Malassezia* (Fig. 2B). Other fungal taxa were present at low relative abundance.

By contrast, the microbial composition of the 2-SD digestate showed a distinct profile. The bacterial community was dominated by *Alcaligenes*, *Proteiniphilum*, *Cloacibacillus*, *Solidesulfobivrio*, *Methanosarcina*, *Methanothrix*, *Methanobacterium*, and

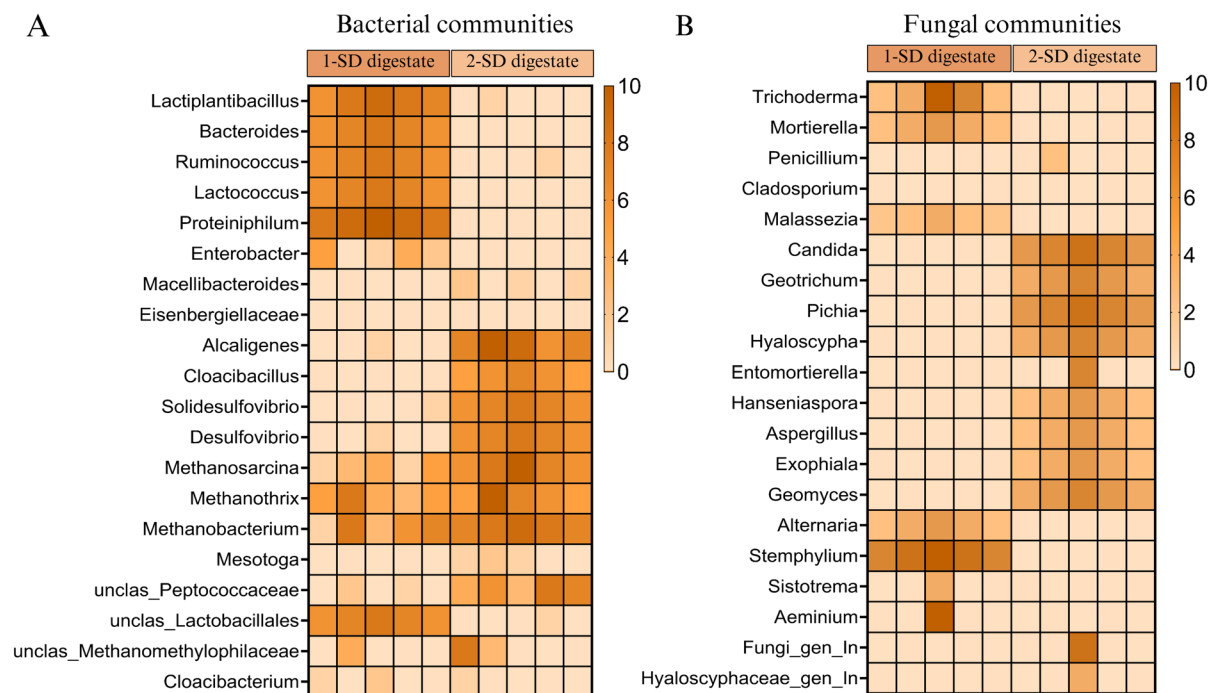


Fig. 2 Heatmap of microbial consortia composition in 1-SD and 2-SD digestates. **A** Shows a heatmap of bacterial communities composition based on relative abundance values obtained from 16S rRNA gene amplicon sequencing. **B** Shows a heatmap of the relative abundance of fungal communities in

both digestates derived from ITS1 and ITS2 amplicon sequencing. Heatmaps were generated from the 20 most abundant taxa. Light yellow indicates the lowest abundance, while brown indicates the highest abundance

Mesotoga, along with a higher relative abundance of unclassified bacterial groups (Fig. 2A). The fungal assemblage in the 2-SD digestate differed markedly from that of the 1-SD digestate and was characterized by a higher relative abundance of *Candida*, *Geotrichum*, *Pichia*, *Hyaloscypha*, *Entomortierella*, and several unclassified fungal taxa (Fig. 2B). Fungal diversity appeared more evenly distributed among taxa in the 2-SD digestate, with no single genus dominating to the same extent as observed in the 1-SD digestate.

Expression of nutrient transporter genes in wheat roots

Utilizing dPCR method to analyze the expression level of genes encoding ion transporters revealed that the expression of the ammonium transporter gene *TaAMT1.2* was significantly higher (Tukey post-hoc test; $p < 0.05$) in both variants fertilized with 1-SD and 2-SD digestates than in the NF control and SF variant (Fig. 3B). Expression level of the reference gene *GAPDH* did not differ significantly among

treatments (Fig. 3A). No statistically significant differences were obtained in the expression of the dual-affinity nitrate transporter gene *TaNRT1.1* nor high-affinity transporter of phosphates—*TaPHT1.10*, among the experimental variants (Fig. 3C–D).

Elemental composition, wheat yield, and biomass production

Wheat yield analysis showed that SF, 1-SD, and 2-SD variants produced statistically significant higher yields than the NF control (Tukey's post-hoc test; $p < 0.001$) and those yield values differ statistically among treatments themselves. The SF variant boosted crop yield by 7.8-fold, while the liquid biofertilizers 1-SD and 2-SD increased wheat yield 2.5-fold and 3.3-fold, respectively, relative to the NF control (Fig. 4A). Straw dry matter was also measured and expressed per unit area. Straw biomass increased 3.9-fold in SF, 1.7-fold in 1-SD, and 2.1-fold in 2-SD relative to the NF control ($p < 0.001$). Both digestate treatments differ from SF significantly, causing

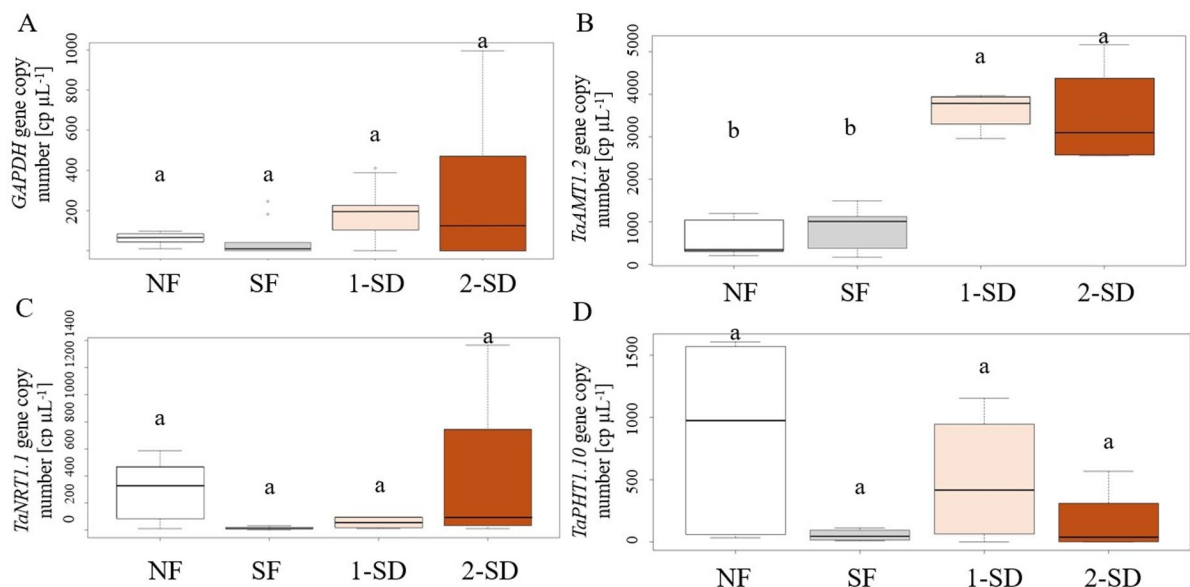


Fig. 3 Expression analysis of ion transporter genes in wheat roots. Boxplots represent median values, shown as central bold line in each box ($n=4-9$). Minimal and maximal values are marked by extended whiskers. White boxes correspond to NF, gray to SF, orange to 1-SD, and brown to 2-SD variants. Statistically significant differences between variants were determined using one-way ANOVA or the Kruskal–Wallis test (for non-normally distributed data), followed by appropriate

post-hoc tests, and are marked in the figure with different statistical letters. Panels demonstrate gene copy number for (A) positive control *GAPDH*, (B) *TaAMT1.2*, (C) *TaNRT1.1*, and (D) *TaPHT1.10* target genes. Gene expression was quantified using digital PCR based on EvaGreen® fluorescence signal absolute quantification and is reported as gene copy number per microliter of sample

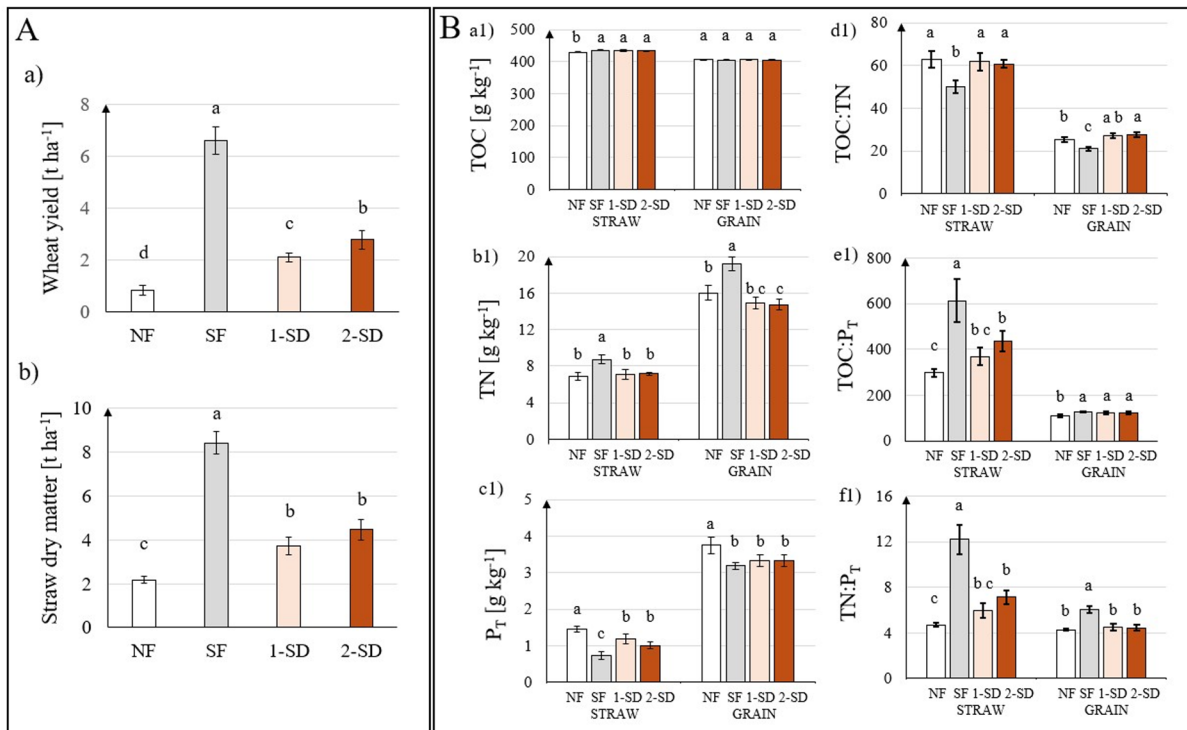


Fig. 4 Crop yield and elemental composition of wheat. In both panels, white bars correspond to NF, gray bars to SF, orange bars to 1-SD, and brown bars to 2-SD treatments. **A** Shows an analysis of wheat yield based on the type of fertilization used. Bars represent (mean ± SD): (a) Wheat yield per area of microcosm, expressed as [t ha⁻¹] and (b) Straw dry matter [t ha⁻¹] ($n=5$). Statistically significant differences between variants are marked with different statistical letters in the figure. **B** Shows the content and ratios of selected ele-

ments (mean ± SD) in cultivated and harvested wheat straw and grains, depending on the type of fertilization ($n=5$). Bars represent mean values, with standard deviation indicated as whiskers. The left side of each panel shows results from wheat straw, while the right side shows results for grain analyses. Analyses are presented in the following order: (a1) TOC, (b1) TN, (c1) P_T, (d1) TOC:TN ratio, (e1) TOC:P_T ratio, and (f1) TN:P_T ratio

less straw dry matter production, but did not differ to themselves (Fig. 4A).

Chemical analyses of selected elements showed that TOC content in straw was slightly higher in all treated variants than in NF control (Tukey's post-hoc test, $p < 0.05$) (Fig. 4B). TN content was significantly higher in the SF variant than in the NF control (Tukey's post-hoc test), with increases of 27.5% in wheat straw and 19.9% in wheat grains ($p < 0.001$). By contrast, TN content in grains was 7.9% lower in 2-SD biofertilized samples than in the NF control ($p < 0.05$) (Fig. 4B). In straw, P_T content was lower in all fertilized variants than in the NF control, with reductions of 50.1% in SF ($p < 0.001$), 18.1% in 1-SD ($p < 0.05$), and 37.7% in 2-SD ($p < 0.001$). Similar trends were observed in grains, where P_T content decreased by

15.0% in SF, 11.0% in 1-SD, and 11.3% in 2-SD compared to NF (Fig. 4B). The TOC:TN ratio in straw was significantly lower only in the SF variant (by 20.5% compared to the NF control; $p < 0.001$). In grains, the TOC:TN ratio was reduced in SF, but increased by 8.3% in the 2-SD variant compared to the NF control ($p < 0.05$). A high TOC:P_T ratio was observed in SF straw, reaching 2.1 times the value of the control ($p < 0.001$). Elevated TOC:P_T ratios were also observed in 2-SD straw and in grains ($p < 0.05$) from all fertilized variants (Fig. 4B). The final ratio evaluated was TN:P_T. In straw, the TN:P_T ratio was 2.6 times higher in the SF variant than in the NF control and 1.5 times higher in the 2-SD fertilized variant than in the NF control ($p < 0.001$). In grains, a significant increase in the TN:P_T ratio was observed only in

the SF variants (Fig. 4B). No significant difference was observed in TOC content among fertilization treatments in grains.

Analysis of factors affecting wheat yield

A linear regression analysis was performed to identify factors influencing crop yield. Selected parameters, including soil nutrient content at harvest and the expression of ion transporter genes in wheat roots, were analyzed in relation to wheat yield across all treatments. The results indicated that soil pH and the expression of the phosphate

ion transporter gene *TaPHT1.10* in wheat roots had a statistically significant slightly positive effect on wheat yield (Fig. 5). According to the regression model, a minor decrease in soil pH was associated with higher wheat yield. Similarly, lower expression levels of *TaPHT1.10* in wheat roots were associated with slender increased crop yield. *TaPHT1.10* acts as phosphorus starvation sensor, the lower expression of this gene may indicate that plants have met the criteria demand for phosphorus in nutrition level and this result may slightly increase crop yield, regarding that R^2 is only 0.3251, which does not stand for strong relationship.

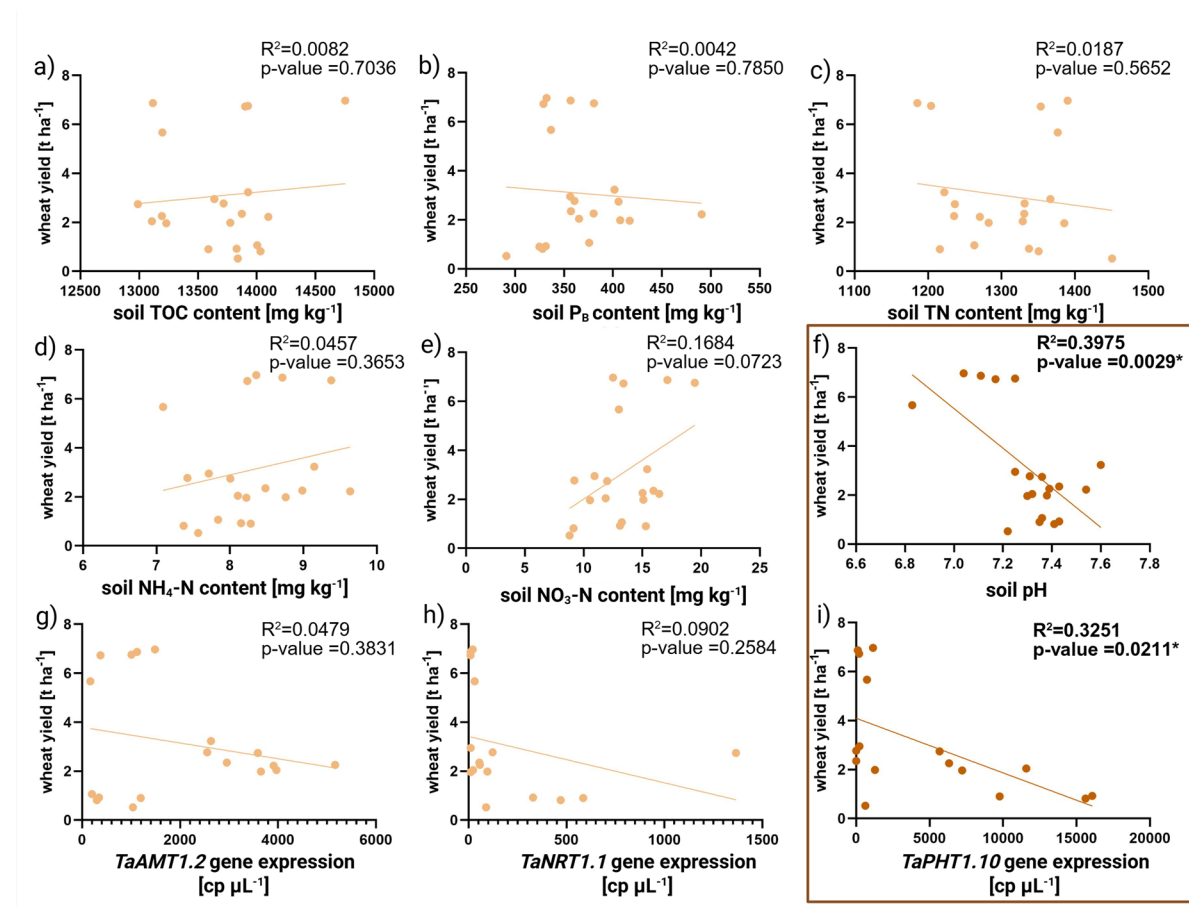


Fig. 5 Linear regression model of factors affecting wheat yield. This figure shows the effect of selected soil parameters at the end of the experiment, including soil elemental composition and pH, as well as ion transporter gene expression in wheat roots, on final wheat yield. For each parameter, p -values and R^2 equations are presented. Statistically significant parameters ($p < 0.05$) are highlighted in bold with brown frames,

and the regression curves are shown in a darker color. Statistically significant values are marked with the asterisk. Analyses are represented in the following order of the parameters: **a** Soil TOC content, **b** Soil P_B content, **c** Soil TN content, **d** Soil NH₄-N content, **e** Soil NO₃-N content, **f** Soil pH, **g** *TaAMT1.2* gene expression, **h** *TaNRT1.1* gene expression, **i** *TaPHT1.10* gene expression

Discussion

Identifying a biological fertilizer (biofertilizer) that can achieve yields comparable to those of synthetically fertilized crops, while also being a by-product consistent with circular economy practices, remains a notable challenge in sustainable agriculture, particularly in wheat cultivation. In this study, we addressed this challenge by directly comparing digestates derived from one-stage (1-SD) and two-stage (2-SD) anaerobic digestion (AD) of molasses when applied as fertilizers under field conditions.

Distinct fertilization properties of one-stage and two-stage digestates

The first hypothesis proposed that digestates derived from different stages of AD differ in their chemical properties and microbial composition. Chemical analyses supported this hypothesis, showing that the 1-SD digestate had higher TN and P_T contents than the 2-SD digestate. However, analysis of nitrogen speciation revealed that the 2-SD digestate contained higher concentrations of mineral nitrogen forms (NH_4-N and NO_3-N) than the 1-SD digestate (Table 3). This result indicates that a larger proportion of nitrogen in the 2-SD digestate exists in forms more readily available to plants. This is further supported by the fact that the application of 2-SD digestate resulted in a higher TOC:TN ratio in wheat grain and a higher TN: P_T ratio in wheat straw (compared to 1-SD digestate), ultimately translating to a 32% higher wheat yield compared to the 1-SD variant ($p < 0.05$). By contrast, nitrogen in the 1-SD digestate was predominantly bound in organic compounds. Nevertheless, it exhibited more balanced TOC:TN and TOC: P_T ratios, which favor microbial mineralization processes and improve the balance between nutrient release and plant uptake. However, analysis of the *TaAMT1.2* gene expression (Fig. 3B) suggests no differences in the plant response between 1-SD and 2-SD digestate treatments. 1-SD and 2-SD treatment led to statistically higher expression of this gene, potentially indicating an enhanced capacity for ammonium ion uptake from the soil, which was still at a high level (Porrás-Murillo et al. 2023). Typically, such upregulation occurs during nitrogen deficiency (Duan et al. 2015; Bajgain et al. 2018). Because soil TN, NO_3-N , and NH_4-N did not differ at harvest (however

tendency toward their higher content in treated soil was maintained), this response may reflect an attempt to maximize nitrogen acquisition under changing late-season conditions. (Bajgain et al. 2018; Wang et al. 2025). Furthermore, the lower nitrogen level in the grain in the 2-SD treatment suggests slight nitrogen deficiency during grain maturing, which may have contributed to the elevated gene expression in that variant, suggesting demand and redistribution of nitrogen for aboveground parts of the plant (Guo et al. 2026). While analyzing fertilizing properties of digestates, the 2-SD digestate contained a less favorable for plants TOC: P_T ratio than the 1-SD digestate. Although TOC inputs were similar in both digestate treatments, this finding indicates less optimal stoichiometry in nutrient content, particularly for microbial nutrient processing, including microbially-mediated phosphorus mobilization and immobilization (Schindlbacher et al. 2011; Lazicki et al. 2020; Luo et al. 2020; Feng et al. 2023; Agostini et al. 2024; Engedal et al. 2025). Moreover, the chemical composition of the 1-SD digestate may reduce the risk of phosphorus immobilization and nitrogen environmental losses, thereby supporting a more sustained and efficient nutrient supply over time. Consistent with the first hypothesis, the digestates also differed in microbial composition. Microbial profiling revealed distinct bacterial and fungal communities in digestates derived from one-stage and two-stage systems. In the 1-SD digestate, the presence of bacterial genera such as *Lactiplantibacillus* and *Lactococcus*, along with a high relative abundance of *Bacteroides*, is particularly relevant in the soil–plant context (Fig. 2A). Species within *Lactiplantibacillus* and *Lactococcus* have been reported in literature data to exhibit plant growth-promoting traits, including phytohormone production and the secretion of biologically active metabolites and peptides, while *Bacteroides*, in turn, contributes to nutrient cycling and phosphorus mobilization in soil (Lidbury et al. 2021; Raman et al. 2022; Zhao et al. 2024; Shah et al. 2025). In our study, the fungal component of the 1-SD digestate, which was dominated by *Trichoderma* (Fig. 2B), was equally important. Other studies demonstrated this genus as widely recognized for its ability to colonize the rhizosphere, stimulate root development and enhance phosphorus solubilization, and uptake efficiency, which is mediated through biochemical activity and modifications to root system architecture in

cereals, including wheat (Sood et al. 2020; Andrzejak et al. 2021; Andrzejak and Janowska 2022; Contreras-Cornejo et al. 2024). The introduction of such taxa into soil via the 1-SD digestate may therefore tend to contribute to localized changes in phosphorus availability. This interpretation is consistent with the observed increase in bioavailable P_B in soil fertilized with 1-SD digestate (Fig. 2B; Table 2). By contrast, the 2-SD digestate was characterized by the presence of taxa belonging to *Alcaligenes* and *Cloacibacillus*, which have been reported in other studies to exhibit plant growth-promoting traits, including non-symbiotic nitrogen fixation, phytohormone production, and indirect enhancement of plant nutrient uptake (Egamberdiyeva 2007; Di Benedetto et al. 2017; Jia et al. 2022), organic matter degradation and enhanced phosphorus mobilization in soil (Luo et al. 2022; Singh et al. 2022; Murindangabo et al. 2023), respectively. However, unlike the 1-SD, the microbial community of the 2-SD digestate in our study was also dominated by taxa such as *Methanosarcina* and *Methanothrix*, which are responsible for methanogenesis in the AD process (Fig. 2B). These microorganisms are not typically associated with soil nutrient cycling or recognized as PGP strains; rather, they are essential for efficient biogas production, the process from which they are derived (Angelidaki et al. 2018; De Vrieze et al. 2018). Their dominance in the 2-SD digestate suggests that, despite higher concentrations of mineral nitrogen forms, the biological potential of the 2-SD digestate to actively modulate soil nutrient availability through microbial interactions may be more limited than that of the 1-SD digestate. Therefore, higher yield in 2-SD digestate treated plant is rather an effect of its mineralization than microbial consortia composition in this treatment.

Changes in soil nutrient availability and plant acquisition strategies

In this study, the only elemental soil chemistry parameter that changed under field conditions was P_B content after treatment with the 1-SD digestate. This finding positively correlates with the previously noted abundance of *Trichoderma* spp. fungi and *Bacteroides* bacteria in the 1-SD digestate, both known from literature data for their phosphorus-solubilizing properties (Sood et al. 2020; Lidbury et al. 2021). No significant differences were observed among

variants in TOC, TN, or concentrations of NH_4 -N and NO_3 -N in the soil. Accordingly, the TOC:TN ratio remained unchanged and was maintained at a relatively low level compared with values reported for other organic fertilizers in the literature (Chuan et al. 2013; Marques da Silva et al. 2024). This finding is relevant from an agroeconomic perspective, as it indicates that, despite the liquid form of digestates, which could promote rapid soil nitrogen transformation or loss under field conditions, they meet wheat demands for mineral nutrition and support effective nutrient utilization without posing risks to agroecosystems, especially soil health (Rose and Wissuwa 2012; Meloni et al. 2024). From a chemical perspective, soil pH is a key parameter in mineral nutrition and fertilization. In this study, soil pH decreased slightly only under the SF treatment (Table 2). This slight reduction in pH may have caused short-time benefits by enhancing ion uptake by the wheat root system, especially bioavailable nutrients in anions forms which require symport with H^+ protons, (Busch and Böttger 1997), which subsequently correlates with the highest straw biomass obtained for this variant (Fig. 5B). However, it needs to be underlined that prolonged use of synthetic mineral fertilizers can cause long-term risk leading to progressive soil acidification, which may affect soil microbiome composition, alter physiochemical properties, and ultimately contribute to soil depletion (Reijnders 2008; Morsetto 2020; Guo et al. 2023; Howe et al. 2024; Salehi et al. 2025). By contrast, both digestates exhibited a more alkaline pH, and their application increased soil pH compared to the SF treatment.

The physiological responses of wheat roots further refine the comparison between the 1-SD and 2-SD digestates and support the second hypothesis that digestate-based fertilization modifies wheat nutrient acquisition mechanisms, particularly by changes in the expression of ion transporters in roots. In this study, several ion transporters relevant to plant nutrition were investigated, enabling evaluation of plant responses to treatment with 1-SD and 2-SD digestates in comparison with NF control and SF treatment at the end of the vegetative season. Both digestate treatments induced a clear increase in the expression of the ammonium transporter gene *TaAMT1.2* relatively to the NF control (Fig. 3A) and SF treatment indicating enhanced reliance on ammonium uptake under digestate fertilization. This response was not observed

in the SF treatment, where nitrogen availability was more stable and did not require strong physiological adjustment. The upregulation of *TaAMT1.2* in both 1-SD and 2-SD treatments occurred despite the absence of differences in soil $\text{NH}_4\text{-N}$ concentrations at the time of harvest (Table 2), suggesting that plants may still actively try to uptake ammonium from the soil to utilize nitrogen efficiently and metabolism at high level (Liu et al. 2015; Dai et al. 2023; Esmaeilzadeh-Salestani et al. 2023). This gene expression could have been upregulated by high ammonium content, demand for nitrogen during grain maturity followed by need for nitrogen redistribution from roots to aboveground parts of the plant, e.g. grains, which was stated above (Ijato et al. 2021; Esmaeilzadeh-Salestani et al. 2023). No significant differences were observed between treatments in the expression of *TaNRT1.1* (Fig. 3C), which acts as a signal sensor, informing the plant about nitrate availability in the soil (Duan et al. 2014; Taulemesse et al. 2015; Fang et al. 2021), together with the fact of unchanged $\text{NO}_3\text{-N}$ content in soil between variants, it may suggest that nitrate was depleted from the root zone prior to sampling, likely due to seasonal changes in soil nitrogen cycling (Fortuna et al. 2003; Taulemesse et al. 2015; Wang et al. 2021). Analysis of the root-specific ion transporter *TaPHT1.10* gene, whose expression is typically induced under phosphorus deficiency (Zhang et al. 2014; Teng et al. 2017), did not reveal a statistically significant change between variants.

Although both digestates elicited similar physiological responses relative to NF and SF, the higher mineral nitrogen content of the 2-SD digestate did not confer a clear physiological advantage over the 1-SD digestate. This finding highlights the need for future research focused on soil-microbiome-plant interactions throughout the vegetative season, when active nutrient uptake occurs, regarding demanding metabolic photosynthesis process.

Determination of wheat yield following digestate-based and synthetic treatments

Comparison of total elemental content (e.g., TN content) suggests that the 2-SD digestate treatment functioned more similarly to SF, consistent with its higher degree of mineralization (Table 2; Fig. 4A). By contrast, analysis of the TOC:TN ratio,

an important determinant of wheat yield, revealed a contrasting tendency that reflects differences in carbon and nitrogen redistribution not only within the digestates but also within treated plants (Dai et al. 2016; Meloni et al. 2024; Li et al. 2022). Another key parameter used to evaluate fertilization status was P_T content, which decreased in all fertilized treatments relative to the NF control in both straw and grains (Fig. 4A). Treatment with SF, 1-SD and 2-SD stimulated more rapid wheat growth, leading to significantly higher overall biomass and higher number of grains (Fig. 4A). If the plant absorbed similar phosphorus level, which can be explained by unchanged *TaPHT1.10* (Fig. 3D) gene expression level across variants the content of P_T per kg of dry matter of straw and grain is smaller in treated plants (Gaj and Górski 2014). Both digestates increased yield relative to the NF control, confirming their fertilizing potential (Fig. 4B). However, the extent of yield improvement differed between 1-SD and 2-SD digestate treatment and for both remained lower than that achieved under SF, reflecting fundamental differences in nutrient availability and retention rather than differences in total nutrient input. Direct comparison of the two digestates indicated that the 2-SD digestate, containing higher concentrations of readily available nitrogen forms, resulted in a proportionally higher yield than the 1-SD digestate, improving the yield for about 32% than in 1-SD treatment. However, taking into consideration the fact that fertilization dose was equal to SF treatment we reported 2.4 times lower yield in 2-SD than in SF, and 3.1 times lower yield in 1-SD than in SF, respectively. This outcome strongly suggests that a substantial fraction of the mineral nitrogen supplied by 1-SD and the 2-SD digestate could not have been retained within the root zone toward the end of the vegetative season. Furthermore, as nitrification occurs seasonally in soil also loses in liquid digestate by natural processes could occur (Fortuna et al. 2003; Wang et al. 2021). This interpretation is consistent with increased *TaAMT1.2* expression under 1-SD and 2-SD digestate treatments. The regression analysis model demonstrated that wheat yield was slightly determined by soil pH. Slight decrease in pH values was positively associated with increased crop yield, however R^2 was around 0.4, indicating that this interaction is not strong (Fig. 5). Notably, only the SF treatment resulted in a slight decrease in soil pH during the experiment (Table 2) and produced the

highest wheat yield, straw biomass, and TN content in both straw and grain. Interestingly, *TaPHT1.10* gene expression in wheat roots also was slightly associated with wheat yield (Fig. 5). Reduced expression of this gene, which acts as an indicator of phosphorus deficiency, may indicate sufficient phosphate uptake revealing that plants have fulfilled nutrient demand for this nutrient, thereby contributing to increased wheat yield. Specifically, SF resulted in the greatest yield increase (approximately seven-fold), while the 1-SD and 2-SD digestate treatments improved crop yield by 2.5-fold and 3.3-fold respectively, relative to the NF control. Linking linear regression model with elemental composition of straw and grains and finally change in *TaAMT1.2* – high-affinity ammonium transporter expression level in roots, these findings strongly support the third hypothesis, that 1-SD and 2-SD digestate treatments modify nitrogen acquisition pattern in wheat cultivated in treated soils, differently than SF treatment, resulting in different crops yields, despite equal nutrient input.

Conclusions

This study demonstrates that 2-SD digestate is a more mineralized biofertilizer, which can be a source of nutrients more easily absorbed by plants, while 1-SD digestate, with a more complex chemical structure and different microbial composition, promotes more balanced nutrient supply leading to enhanced phosphorus availability in soil. Higher mineralization in 2-SD digestate was associated with proportionally higher wheat yield than the control and 1-SD digestate treatment. However, because both digestates were applied in liquid form, yields were lower than those obtained with SF. Treatments with 1-SD and 2-SD digestate despite initial similar inputs of nitrogen to this in the synthetically fertilized variant led to changes in nitrogen acquisition patterns in wheat roots, connected with plants mechanisms to utilize ammonium sources in soil. Therefore, further studies, especially during the vegetative season, when plants absorb nutrients more actively and perform effective photosynthesis and anabolic processes, are needed to evaluate more detailed physiological responses in wheat. Also, these analyses together with results demonstrated in this work may be beneficial for finding more stable digestate forms or more adjusted

strategies to avoid nutrient leaching risk from digestate-based fertilization.

Author contributions Marta Galas: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Visualization, Writing- Original Draft; Anna Detman-Ignatowska: Conceptualization, Methodology, Validation, Investigation, Writing—Review & Editing; Jerzy Jonczak: Conceptualization, Methodology, Validation, Investigation, Writing—Review & Editing; Joanna Szewińska: Validation, Writing—Review & Editing; Aleksandra Chojnacka: Conceptualization, Methodology, Validation, Supervision, Funding acquisition, Writing—Review & Editing.

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Data availability Data sets for this study is available in open data repository doi: (<https://doi.org/10.18150/7ZYVRV>).

Declarations

Conflicts of interests The authors declare no conflict nor competing of interest.

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